ROBO AI

IoT-Based Weather Station

ESP32 Source Code

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```

#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <HTTPClient.h>
#include "DHT.h"

// -------- DHT22 CONFIGURATION --------
#define DHTPIN 15
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);

// -------- WIFI CONFIGURATION --------
const char* WIFI_SSID = "Wokwi-GUEST";
const char* WIFI_PASSWORD = "";

// -------- THINGSPEAK CONFIGURATION --------
// Replace with your private ThingSpeak Write API Key
const char* WRITE_API_KEY = "##############";

const char* SERVER_URL =
    "https://api.thingspeak.com/update";

// Upload interval: 20 seconds
const unsigned long UPLOAD_INTERVAL = 20000;
unsigned long lastUploadTime = 0;

void connectWiFi() {

  Serial.print("Connecting to WiFi");

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD, 6);

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println();
  Serial.println("WiFi Connected!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
}

void uploadToThingSpeak(float temperature,
                        float humidity) {

  WiFiClientSecure client;
  client.setInsecure();
```

```
50
51    HTTPClient http;
52
53    Serial.println("Uploading data to ThingSpeak...");
54
55    if (http.begin(client, SERVER_URL)) {
56
57      http.addHeader("Content-Type",
58                     "application/x-www-form-urlencoded");
59
60      String postData = "api_key=" + String(WRITE_API_KEY);
61      postData += "&field1=" + String(temperature, 2);
62      postData += "&field2=" + String(humidity, 2);
63
64      int httpResponseCode = http.POST(postData);
65
66      if (httpResponseCode > 0) {
67
68        String response = http.getString();
69
70        Serial.print("HTTP Response Code: ");
71        Serial.println(httpResponseCode);
72
73        Serial.print("ThingSpeak Entry ID: ");
74        Serial.println(response);
75
76        if (response.toInt() > 0) {
77          Serial.println("SUCCESS: Data uploaded!");
78        } else {
79          Serial.println(
80            "Upload not accepted. Check API key and channel."
81          );
82        }
83
84      } else {
85
86        Serial.print("Upload Error: ");
87        Serial.println(httpResponseCode);
88
89      }
90
91      http.end();
92
93    } else {
94
95      Serial.println("Unable to connect to ThingSpeak!");
96
97    }
98  }
99
100 // -------- SETUP --------
101 void setup() {
102
103   Serial.begin(115200);
104
105   Serial.println();
106   Serial.println("================================");
107   Serial.println("    IoT WEATHER STATION");
```

```
  Serial.println("================================");

  dht.begin();

  Serial.println("DHT22 Sensor Initialized!");

  connectWiFi();

  Serial.println("System Ready!");
}

// -------- MAIN LOOP --------
void loop() {

  if (WiFi.status() != WL_CONNECTED) {
    connectWiFi();
  }

  if (millis() - lastUploadTime >= UPLOAD_INTERVAL ||
      lastUploadTime == 0) {

    lastUploadTime = millis();

    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();

    if (isnan(temperature) || isnan(humidity)) {

      Serial.println("ERROR: Sensor reading failed!");

      return;
    }

    Serial.println();
    Serial.println("--------------------------------");
    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.println(" C");

    Serial.print("Humidity: ");
    Serial.print(humidity);
    Serial.println(" %");

    Serial.println("--------------------------------");

    uploadToThingSpeak(temperature, humidity);
  }

  delay(100);
}
```

Listing 1: ESP32 firmware for DHT22 and ThingSpeak upload